

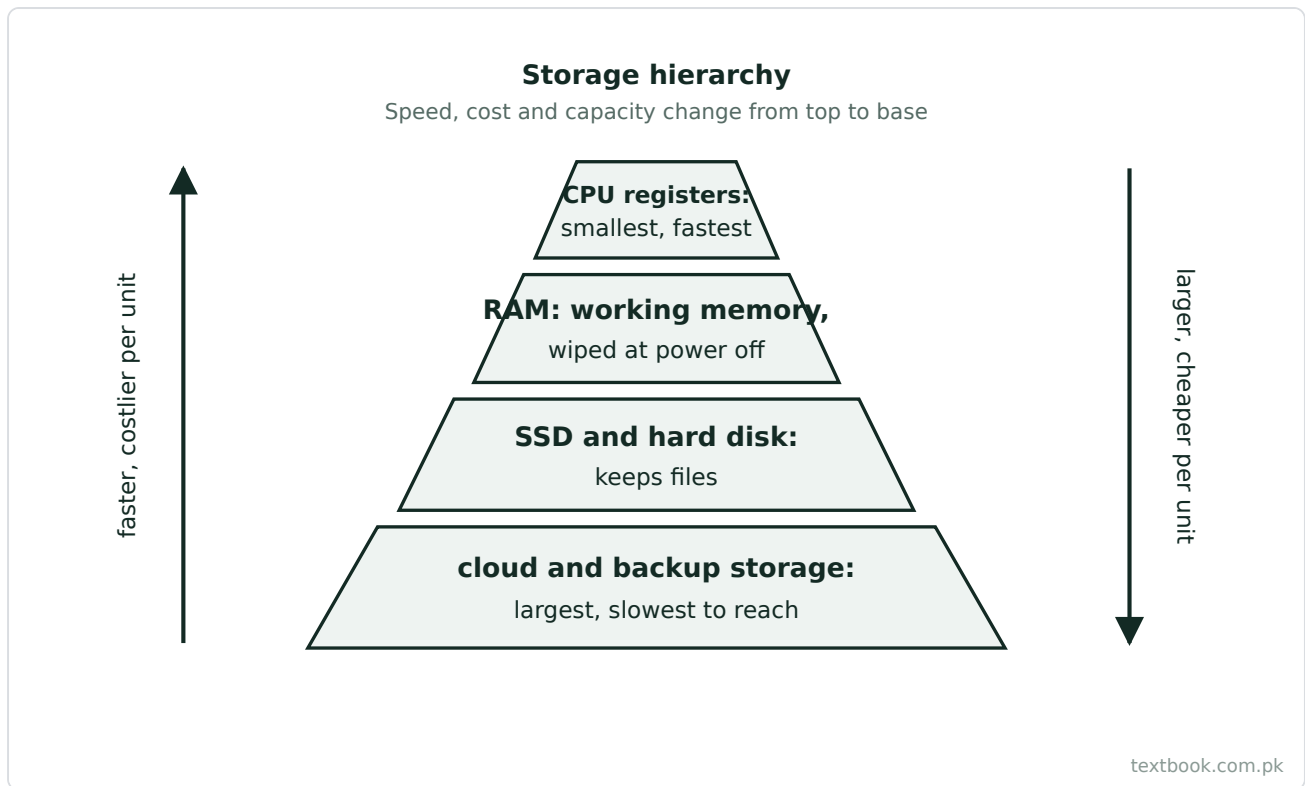
Draft - expert review pending

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Topic 2.2 - Memory inside and outside the computer

A real classroom situation

Miss Kiran is typing the class results into a spreadsheet when the power fails, as it often does in Larkana in summer. The lights return after a minute. She switches the computer back on, opens the spreadsheet, and finds only the version she saved yesterday. An hour of typing is gone. Her cooperating teacher is not surprised: "The computer only remembers what you save," she says. "While you work, it holds everything in a kind of whiteboard that wipes itself clean when the power goes." That whiteboard is primary memory, and understanding it and its stubborn cousin, storage, is what this topic is about.



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Explanation

Primary memory: RAM and ROM

Primary memory is the memory the computer uses while it works. It has two main forms. RAM (Random Access Memory) is the working memory into which programs and data load while you use them. Bourgeois and colleagues (2019) describe its two key traits: it can transfer data much faster than the hard disk, and it is volatile, meaning its contents are lost when the power goes off. That is the whiteboard from the classroom story. More RAM lets a computer keep more programs and bigger files open comfortably; when people say a computer is "slow with many tabs open", RAM is usually what they mean.

ROM (Read Only Memory) is the other form of primary memory. It holds the small, permanent instructions the computer needs to start itself. ROM is non-volatile: its contents survive power off, and normally they are not meant to be rewritten in everyday use. You do not choose ROM or manage it; you simply rely on it every time the machine boots.

Secondary storage and the storage hierarchy

Secondary storage is memory that keeps data whether or not the power is on: the hard disk, the solid state drive, the flash drive, the DVD, the memory card. A hard disk is non-volatile storage (Bourgeois et al., 2019); an SSD performs the same job with no moving parts, which makes it faster, lighter and more reliable, at a higher price for the same capacity. This is where your saved spreadsheet lives, and why yesterday's copy survived the power cut while today's typing did not.

The two kinds of memory are not rivals; they are a team, and they form a ladder called the storage hierarchy. At the top sit the processor's own tiny, extremely fast registers. Just below is RAM: fast, moderate in size, and forgetful. Below that is the SSD or hard disk: large, affordable, and permanent. At the bottom sits everything you keep far away but reach through a network: the school server down the corridor, the cloud drive on the internet. Each step down the ladder buys you more space for less money, at the cost of speed. That single trade explains the shape of every computer you will meet.

RAM, ROM and secondary storage: a comparison

Property	RAM	ROM	Secondary storage
What it holds	programs and data while in use	start-up instructions	saved files
Survives power off	✗ no	✓ yes	✓ yes
Speed	fastest of the three	fast	slower
Can the user add more	yes	not normally	yes
Typical size in a school PC	4 to 16 GB	a few MB	256 GB to 2 TB

Read each row across to compare the three types of computer memory and storage.

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The everyday proof of the hierarchy is the save habit itself. While Kiran types, her work lives in RAM. The moment she presses save, a copy is written to the

disk. Until that press, the disk does not have it. A teacher who understands this teaches pupils to save early and often, and to treat an unsaved document as a loan, not a possession.

A second practical lesson follows for school purchases. When a computer feels slow, the fix depends on which rung is exhausted: too little RAM shows as slowness when many programs are open; a full or old disk shows as slowness in everything, including start-up. Knowing which memory is which turns "buy a new computer" into a precise question: which rung of the ladder is the bottleneck?

Activity: The power-cut drill

Work in pairs. About 15 minutes.

1. Partner A describes a task being done on a computer (for example: editing the school magazine cover in a photo tool, with the file last saved 40 minutes ago). Partner B announces "power cut!".
2. Together, state exactly what is lost, what survives, and where the surviving copy lives (RAM, ROM, disk, or cloud). Use the hierarchy words.
3. Swap roles with a new task. Cover at least four tasks between you, including one where the file was saved to a cloud drive and one where the computer was only starting up.
4. End by writing one sentence: the rule a pupil should be taught about saving.

Check your understanding

1. Define RAM and state its two key properties in one sentence. (*Remember*)
2. Explain why a document that was never saved disappears after a power cut, using the words RAM and secondary storage. (*Understand*)
3. A school computer takes ten minutes to start and is slow even with one program open, while another starts quickly but freezes when five programs run. Say which rung of the storage hierarchy is the likely bottleneck in each machine, and why. (*Apply*)
4. Name one thing ROM does that RAM cannot. (*Remember*)

Summary

Primary memory is the computer's working memory: RAM, which is fast, volatile and holds whatever is in use; ROM, which is small, permanent and starts the machine. Secondary storage (hard disk, SSD, flash drive, DVD) keeps files across power cuts. Together they form a hierarchy trading speed for size at every step, and the save habit exists precisely because RAM forgets and disks remember.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define RAM, ROM and secondary storage and state what each holds.
- ☐ I can explain what is lost and what survives a power cut in a described scenario.
- ☐ I can read the storage hierarchy and say what each rung trades away.

Try this at your practicum school

Find out how much RAM and how much storage the lab or office computers have (ask the in-charge, or look in the system information screen). Note the numbers, and ask one question: which rung do they run out of first in daily use? Bring the answer back.

Summative task

A school wants to keep end-of-term results for ten years and also needs its office computer to handle the accounts software comfortably. In one page, recommend what the school should spend on (more RAM, an SSD upgrade, or an external drive plus a cloud account), naming for each purchase which rung of the hierarchy it serves and what problem it solves.

Marking guide (10 marks):

- Each purchase correctly mapped to its rung of the hierarchy: 3 marks.

- Each problem solved is the right problem for that rung (Analyse): 4 marks.
- The ten-year-keeping requirement and the comfort requirement are both addressed: 2 marks.
- Clear writing within one page: 1 mark.

Further reading

- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 2. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- Norton, P. *Introduction to Computers*. McGraw-Hill Education. (Course guide reading; print monograph, cited at bibliographic level.)
- Khan Academy. *Computers and the Internet*.
<https://www.khanacademy.org/computing/computers-and-the-internet>

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